

## Problem 1: International Trade

The following table shows how many cellphones or apples that a person in Country A or Country B can produce per day. For simplicity assume that all workers within each country have the same skills and that they can move freely between industries within each country, but they cannot move across countries. Furthermore, assume the Country A has 100 workers and Country B has 90 workers.

|           | <b>Output per person per day</b> |               |
|-----------|----------------------------------|---------------|
|           | <b>Cellphones</b>                | <b>Apples</b> |
| Country A | 4                                | 20            |
| Country B | 1                                | 3             |

Use this information to answer the following questions on the next slides.



## Problem 1: Questions (a)-(b)

- (a) Draw the production possibilities frontier (PPF) for each country. Your PPF will show the daily production possibilities for each country. Assume constant opportunity costs (OC).
- (b) Which country has the absolute advantage in producing cellphones? Or in producing apples?



## Problem 1: Questions (c)-(e)

- (c) Calculate the OC of producing a cellphone in each country.
- (d) Calculate the OC of producing an apple in each country.
- (e) According to the law of comparative advantage, which product will the Country A export? Which product will the Country B export?



## Problem 1: Questions (f)-(g)

- (f) Assume the countries decide to trade with each other. Each specializes in the production of the good in which they have the comparative advantage and exports the good in which they have the comparative advantage. The terms of trade between the two is set at 20 cellphones for 200 apples. After they finish trading, how many cellphones and how many apples will each country be able to consume?
- (g) Given your answer in part f), will each country's consumption point after trade be inside its PPF? Outside its PPF? Or on its PPF? Provide an explanation.



## Problem 2: Two Carpenters

Suppose Carpenter A and B both produce Chairs and Desks. The following table shows combinations of the goods that each of them can produce in a day.

| Carpenter A |        | Carpenter B |        |
|-------------|--------|-------------|--------|
| Desks       | Chairs | Desks       | Chairs |
| 0           | 10     | 0           | 20     |
| 1           | 8      | 1           | 16     |
| 2           | 6      | 2           | 12     |
| 3           | 4      | 3           | 8      |
| 4           | 2      | 4           | 0      |
| 5           | 0      |             |        |



## Problem 2: Questions (a)-(c)

- (a) Who has an absolute advantage in the production of chairs? What about desks? Explain why.
- (b) Calculate the opportunity cost of producing desk in each carpenter. Based on your calculations, which carpenter has a comparative advantage in desk production?
- (c) Calculate the opportunity cost of producing chair in each carpenter. Based on your calculations, which carpenter has a comparative advantage in chair production?



## Problem 2: Questions (d)

- (d) For each of the following terms of trade, determine which carpenter – if any – would agree to trade under these terms.
- ▶ 3 desks: 10 chairs
  - ▶ 2 desks: 7 chairs
  - ▶ 1 desks: 1 chairs



## Problem 2: Question (e)

- (e) Suppose that Carpenter A is currently producing 2 desks and 6 chairs, and Carpenter B is producing 3 desks and 8 chairs. Explain why carpenters could both be better off (i.e., consume more) if they specialize in producing only one good and then engage in trade.





As an extension of those exercises, think about how we can relate above exercises in terms of Marginal Utility, Marginal Rate of Substitution, and Indifference Curve:

- ▶ Marginal Utility vs Marginal Productivity
- ▶ Marginal rate of Substitution vs Marginal rate of transformation
- ▶ Indifference Curve vs PPF

